Dynamic Risk Surface Modelling of Cytopenia Recovery and Infection Vulnerability During Bispecific Antibody Therapy in High-Risk Lymphoma

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ABSTRACT

Cytopenia, immune impairment, and infection vulnerability during bispecific-antibody therapy evolve on different time scales. We developed and audited a literature-parameterized dynamic risk surface model that explicitly separates literature supported biological context, provisional structural coefficients, and model-based outputs. Seven continuous states represent marrow reserve $R(t)$, neutrophil reserve $N(t)$, platelet reserve $P(t)$, immune competence $G(t)$, inflammatory burden $C(t)$, microbial pressure $B(t)$, and cumulative infection hazard $H(t)$. Treatment is represented by a normalized treatment-associated stress forcing $E(t)$, rather than drug concentration or a validated pharmacokinetic profile. The executable model combines nonlinear ordinary differential equations, a computational stability coordinate, three state latent regime dynamics, global and local sensitivity analyses, stochastic uncertainty propagation, scenario stress testing, numerical robustness assessment, synthetic mechanism recovery comparison, and structural exposure schedule sensitivity. First passage and sequential decision formulations provide additional mathematical structure for future calibrated analyses. In the reference configuration, neutrophil reserve reached a nadir of 0.724 on day 74.5, minimum stability was 0.708 on day 74.6, and the 84-day model-based infection risk output was 34.2%. The final latent state distribution was 34.5% controlled, 50.2% strained, and 15.4% overloaded. Under the high frailty scenario, model-based risk increased to 49.2%, whereas neutrophil nadir changed only from 0.724 to 0.719, illustrating an assumption-dependent separation between cytopenia depth and broader infection vulnerability. Across 500 stochastic perturbations, median model-based risk was 34.9% (2.5^{th}-97.5^{th} percentile, 27.3%-42.2%). The model provides a mathematically explicit and reproducible hypothesis-based platform for interrogating dynamic vulnerability during bispecific antibody therapy. Its numerical outputs are not clinical incidences, treatment effects, or validated decision thresholds, patient-level calibration and external validation are required before clinical interpretation or decision use.

Keywords: Bispecific Antibody, Cytopenia, Dynamic Risk Surface, Infection Vulnerability, Mathematical Modelling, Regime Switching.

1. INTRODUCTION

CD20×CD3 bispecific antibodies have expanded therapeutic options for patients with relapsed or refractory B-cell lymphomas. Clinical studies of epcoritamab, glofitamab, and mosunetuzumab have shown substantial therapeutic activity while also documenting clinically relevant hematologic, immune, inflammatory, and infectious complications in heavily pretreated populations (Thieblemont et al., 2023, Dickinson et al., 2022 and Budde et al., 2022). Infection focused evidence further indicates that infectious complications constitute an important component of toxicity during bispecific antibody therapy (Reynolds et al., 2024). These observations motivate consideration of multiple interacting processes including marrow suppression, neutrophil depletion and recovery, immune impairment, inflammatory activity, host vulnerability, treatment exposure, and supportive care, rather than infection risk as the consequence of a single contemporaneous laboratory measurement.

A mathematical gap nevertheless remains between such clinical observations and their representation as a time dependent system. Reported event frequencies, toxicity grades, and measurements obtained at discrete clinical visits provide important empirical summaries, but they do not

by themselves specify how interacting biological states evolve continuously or how cumulative vulnerability emerges from their joint trajectories. Coupled ordinary differential equations provide a natural model for representing interacting biological variables as continuously evolving dynamical states (Daun et al., 2008). Two trajectories may therefore reach similar neutrophil nadirs while differing in immune competence, inflammatory burden, microbial pressure, host frailty, or duration of impaired stability. The unresolved question is whether similar cytopenia depths necessarily imply similar cumulative vulnerability histories. Addressing this question requires a dynamic rather than purely cross-sectional representation. In the present study, published clinical evidence is used to motivate biological structure and provide plausibility context, but reported clinical event rates are deliberately not converted into unverified patient specific kinetic coefficients.

To address this gap, we formulate seven state nonlinear ODE equations representing marrow reserve $R(t)$, neutrophil reserve $N(t)$, platelet reserve $P(t)$, immune competence $G(t)$, inflammatory burden $C(t)$, microbial pressure $B(t)$, and cumulative infection hazard $H(t)$. ODE-based models provide an established model for representing interacting biological processes evolving continuously in time (Daun et al., 2008), while continuous time multistate models provide a complementary representation of transitions among discrete or latent biological states (Zingoni et al., 2021). Because conclusions from a literature-parameterized model may depend strongly on provisional coefficients, the model combines a computational stability coordinate and three-state CTMC with Latin-hypercube PRCC global sensitivity analysis, local elasticity, stochastic uncertainty propagation, scenario stress testing, and numerical robustness assessment (Renardy et al., 2021). A synthetic mechanism recovery experiment is used only as an internal structural recovery check. First passage geometry and a formal sequential decision formulation provide additional mathematical states for future analyses, consistent with mathematical oncology decision models, clinical optimization would require adequately calibrated and validated state dynamics (Jarrett et al., 2020). The present executable pipeline therefore does not claim an optimized intervention time or clinical treatment policy.

A further modelling challenge concerns treatment exposure itself. A mathematical forcing schedule can be misinterpreted as a drug specific dosing or pharmacokinetic profile when its parameters have not been estimated from pharmacokinetic data. To avoid this interpretation, the present model represents treatment through a normalized, dimensionless treatment associated stress forcing $E(t)$, rather than serum drug concentration, receptor occupancy, or validated pharmacokinetic exposure. Its amplitudes, decay coefficient, and timing are explicitly treated as provisional structural assumptions, and alternative schedules are examined to quantify the dependence of model trajectories on this choice. Accordingly, the objective of this study is to determine whether a dynamic reserve stress hazard system can make and interrogate time dependent separation between cytopenia depth and broader infection vulnerability, while identifying the structural assumptions and parameters that most strongly govern this behavior. The model is intended as a reproducible mathematical proof-of-concept for hypothesis generation and future observed data calibration, rather than a validated bedside risk score, drug specific pharmacokinetic model, or clinical decision rule.

2. METHODS

2.1. Study Design

This study was designed as a literature-parameterized theoretical and computational proof-of-concept of dynamic cytopenia recovery and infection vulnerability during bispecific antibody therapy. Published clinical studies were used to motivate the biological states and to provide external plausibility context for treatment associated cytopenia, immune impairment, and infection vulnerability (Thieblemont et al., 2023 and Reynolds et al., 2024). Clinical event rates were not converted into patient specific kinetic coefficients, and no patient level data were used for model fitting or validation.

Three categories of quantities were distinguished throughout the analysis: (i) clinical observations reported in the literature, used only as biological context, (ii) provisional structural coefficients defining the computational experiments, and (iii) model-based outputs obtained from the equations and declared parameter values. Accordingly, simulated infection risk values, stability boundaries, latent regime probabilities, and scenario differences were interpreted as mathematical outputs rather than clinical incidence estimates, validated thresholds, or treatment effects.

The computational workflow consisted of a reference simulation, prespecified scenario analyses, global and local sensitivity analyses, stochastic uncertainty propagation, numerical robustness assessment, synthetic mechanism recovery comparison, and treatment stress schedule sensitivity. This design was intended to examine the internal behaviour and assumptions of the proposed dynamical system rather than to establish clinical predictive performance.

2.2. Dynamic Reserve and Hazard Model

Seven continuous state variables were used to represent marrow reserve $R(t)$, neutrophil reserve $N(t)$, platelet reserve $P(t)$, immune competence $G(t)$, inflammatory burden $C(t)$, microbial pressure $B(t)$, and cumulative infection hazard $H(t)$. Coupled ordinary differential equations provide a standard model for representing interacting biological processes evolving continuously over time (Daun et al., 2008). The implemented system was

$$\frac{dR}{dt} = \rho_R(1 - R) - \alpha_E E(t)R - \alpha_C CR,$$

$$\frac{dN}{dt} = k_N R(1 + u_G) - d_N N - q_N E(t)N,$$

$$\frac{dP}{dt} = k_P R - d_P P - q_P E(t)P,$$

$$\frac{dG}{dt} = k_G R - d_G G - q_G E(t)G,$$

$$\frac{dC}{dt} = c_E E(t) + c_B B - d_C C,$$

$$\frac{dB}{dt} = \lambda_B(1+F) - \left[\frac{\text{clear}_B NG}{K_{\text{clear}} + NG} + u_P\right]B,$$

$$\frac{dH}{dt} = h_0 \exp\{\beta_N[1-N]_+ + \beta_G[1-G]_+ + \beta_B B + \beta_C C + \beta_F F\},$$

where $[z]_+ = \max(z, 0)$, F denotes the model frailty parameter, and the remaining coefficients define the provisional reserve, injury, recovery, microbial, and hazard relationships. Cumulative model-based infection risk was calculated as

$$P_{\text{inf}}(t) = 1 - \exp[-H(t)].$$

This transformation follows the standard relationship between cumulative hazard and event probability in time-to-event analysis (Kalbfleisch and Prentice, 2002). In the present study, however, P_{inf} is a model-based risk output, because $H(t)$ was not estimated from observed patient infection times.

2.3.Treatment Stress Forcing

Treatment exposure was represented by a dimensionless treatment-associated stress forcing,

$$E(t) = \sum_j A_j\, I\big(t \geq t_j\big)\exp\big[-k_E\big(t - t_j\big)\big].$$

Importantly, $E(t)$ was not interpreted as serum drug concentration, receptor occupancy, or a validated pharmacokinetic exposure. It was introduced only as a structural forcing term that allows treatment associated stress to enter the reserve equations in a transparent and reproducible manner.

The reference configuration used amplitudes of 0.20, 0.35, and 0.50 at days 0, 7, and 14, followed by amplitude 0.55 at days 28, 42, 56, and 70, with $k_E = 0.35\ \text{day}^{-1}$. These values were provisional modelling assumptions rather than estimates of clinical dose intensity or drug half-life.

Because the resulting trajectories can depend on the assumed forcing schedule, this dependence was examined explicitly rather than treated as fixed biological knowledge. Two alternative structural schedules, weekly after step-up and 21-day after step-up forcing, were compared with the reference configuration. These analyses were interpreted as structural sensitivity experiments, not comparisons of clinical treatment regimens.

2.4. Stability Coordinate, First-Passage Geometry, and Latent Regimes

A computational stability coordinate was defined from the normalized biological states as

$$S(t) = 0.40R + 0.25N + 0.15P + 0.20G - 0.12C - 0.10B$$

This quantity was used as a composite mathematical summary of reserve preservation relative to inflammatory and microbial burden. The weights were provisional structural coefficients and were not estimated from patient outcomes. For descriptive interrogation of the trajectory, stability drawdown was defined as

$$D(t) = \max\{0, S_{\text{ref}} - S(t)\}.$$

The values $S = 0.72$ and $S = 0.62$ were used only as computational reference boundaries. They were not clinical cut-offs and were not used to classify actual patients. A three state continuous time Markov representation was additionally used to summarize the trajectory as controlled, strained, or overloaded:

$$\frac{dp}{dt} = pQ[S(t)], \qquad p_C + p_S + p_O = 1$$

Continuous time multistate models provide a basic model for describing transitions among discrete states over time (Jackson, 2011). Here, the transition intensities were provisional functions of the computational stability coordinate rather than quantities estimated from observed clinical transitions. In the executable implementation, regime probabilities were advanced numerically on the output grid using an explicit Euler update followed by clipping and renormalization. The resulting probabilities were therefore interpreted as numerical latent state summaries, rather than exact CTMC transition probabilities or clinically estimated state prevalence.

2.5. Reference Simulation and Scenario Analysis

The continuous-state system was simulated over an 84-day horizon using the adaptive RK45 algorithm, relative tolerance 10^{-8}, absolute tolerance 10^{-10}, and a 0.1-day output grid. Adaptive Runge-Kutta methods are standard numerical approaches for non-stiff initial value problems (Dormand and Prince, 1980 and Virtanen et al., 2020). The reference initial state was

$$(R, N, P, G, C, B, H) = (0.92, 0.95, 0.94, 0.92, 0.08, 0.07, 0).$$

All state values except cumulative hazard were normalized computational quantities rather than direct laboratory measurements. Prespecified scenarios were used to examine whether changes in selected structural assumptions produced interpretable differences from the reference trajectory. The scenarios represented lower relative dose intensity, growth factor support, antimicrobial prophylaxis, combined adaptation, higher frailty, increased marrow injury, and increased neutrophil suppression.

Scenario comparisons were descriptive computational perturbations. They were not intended to estimate causal treatment effects or expected clinical benefits of supportive-care interventions. A two dimensional vulnerability surface was also conducted by varying frailty from 0.10 to 0.90 and relative dose intensity from 0.70 to 1.20 while retaining the remaining reference assumptions. The resulting surface therefore represents model behaviour over a declared parameter grid rather than an empirically estimated clinical risk surface.

2.6. Uncertainty and Robustness Analysis

Parameter dependence was evaluated using complementary global and local approaches. Global sensitivity analysis used 600 Latin hypercube samples across ten prespecified parameter ranges. Latin hypercube sampling provides a stratified approach for exploring multidimensional parameter spaces (McKay et al., 1979). Partial rank correlation coefficients (PRCCs) were then calculated for three model

outputs: 84-day model-based infection risk, minimum stability, and neutrophil nadir. PRCC is commonly used to assess monotonic parameter output relationships in nonlinear biological models (Renardy et al., 2021).

Local sensitivity was assessed using symmetric $\pm 1\%$ perturbations around the reference parameter values and reported as local elasticities. These analyses quantify the behaviour of the model around its declared parameterization and do not establish parameter identifiability from clinical data. Uncertainty propagation was examined using 500 stochastic perturbation runs in which $\alpha_E, q_N, q_G, \lambda_B, h_0$, frailty, and relative dose intensity were varied around the reference configuration according to the distributions specified in the computational implementation. The resulting distribution of model-based infection risk was summarized using the median and 2.5^{th}-97.5^{th} percentiles. These intervals represent simulation uncertainty under the declared perturbation scheme, not clinical confidence or prediction intervals. Numerical robustness was evaluated by repeating the reference calculation under coarse, reference, and tighter solver-grid settings. Agreement across these configurations was used to determine whether the reported reference outputs were materially dependent on numerical resolution.

2.7. Synthetic Mechanism Recovery Analysis

A limited synthetic comparison was performed as an internal computational check. Sparse pseudo observations were sampled every seven days from the model-based reference neutrophil trajectory and Gaussian noise with standard deviation 0.018 was added. The pseudo observations were compared with three candidate representations, the dynamic reserve model, a 28-day harmonic curve, and a bi-exponential phenomenological model. Model comparison used Akaike and Bayesian information criteria (AIC and BIC), which penalize goodness of fit according to model complexity (Akaike, 1974 and Schwarz, 1978). The dynamic reserve model estimated two multiplicative parameters in this synthetic comparison. Because the pseudo observations were conducted from the mechanistic reference trajectory itself, this analysis was interpreted only as an internal mechanism recovery test. It does not use superior predictive performance on clinical data.

2.8. Mathematical Interpretation

The implemented equations were constructed on a finite 84-day horizon with bounded treatment forcing and finite coefficients. The continuous state simulations therefore provide a numerically tractable computational system under the declared initial conditions and parameterization. This mathematical tractability is not interpreted as evidence that the parameter values are biologically identified or clinically valid. Structural and practical parameter identifiability were not established in the present study. Likewise, the current executable analysis does not estimate patient specific parameters, solve an optimal control policy, or identify an optimal intervention time. A Bellman type sequential decision formulation can be defined for future calibrated versions of the model,

$$V(t,x,p) = \min_a\{\, c(a,x) + \mathbb{E}\big[V\big(t+\Delta t, X(t+\Delta t), p(t+\Delta\, t)\big)\big]\},$$

but this recursion was not solved in the present computational analysis and produced no results reported in this study. It is therefore included only to define a possible future mathematical extension, consistent

with the use of calibrated dynamical models for treatment decision analysis (Jarrett et al., 2020). Patient level inference would require longitudinal observed data, parameter-identifiability assessment, estimation of patient or population level parameters, and independent validation.

3. RESULTS

3.1. Reference Model Dynamics

Under the declared reference parameterization, the 84-day simulation remained numerically finite. The normalized treatment associated stress forcing made repeated transient perturbations in the reserve trajectories rather than a smooth monotonic decline (Figure 1). Neutrophil reserve reached a nadir of 0.724 on day 74.5. The corresponding nadirs of marrow reserve, platelet reserve, and immune competence were 0.773, 0.769, and 0.761, respectively. Minimum computational stability was 0.708 on day 74.6. The trajectory spent 10.2 model-days below the provisional $S = 0.72$ computational boundary and did not cross the $S = 0.62$ boundary. The 84-day model-based cumulative infection risk output was 0.342 (34.2%). These quantities are outputs of the specified equations and parameterization and are not observed clinical incidences or validated clinical thresholds.

The latent regime calculation showed a progressive redistribution from the controlled state toward the strained and overloaded states over the simulated horizon (Figure 1). At day 84, the model-based state probabilities were 0.345 controlled, 0.502 strained, and 0.154 overloaded. These probabilities summarize the model trajectory under the provisional CTMC formulation. They are not estimates of clinical population prevalence.

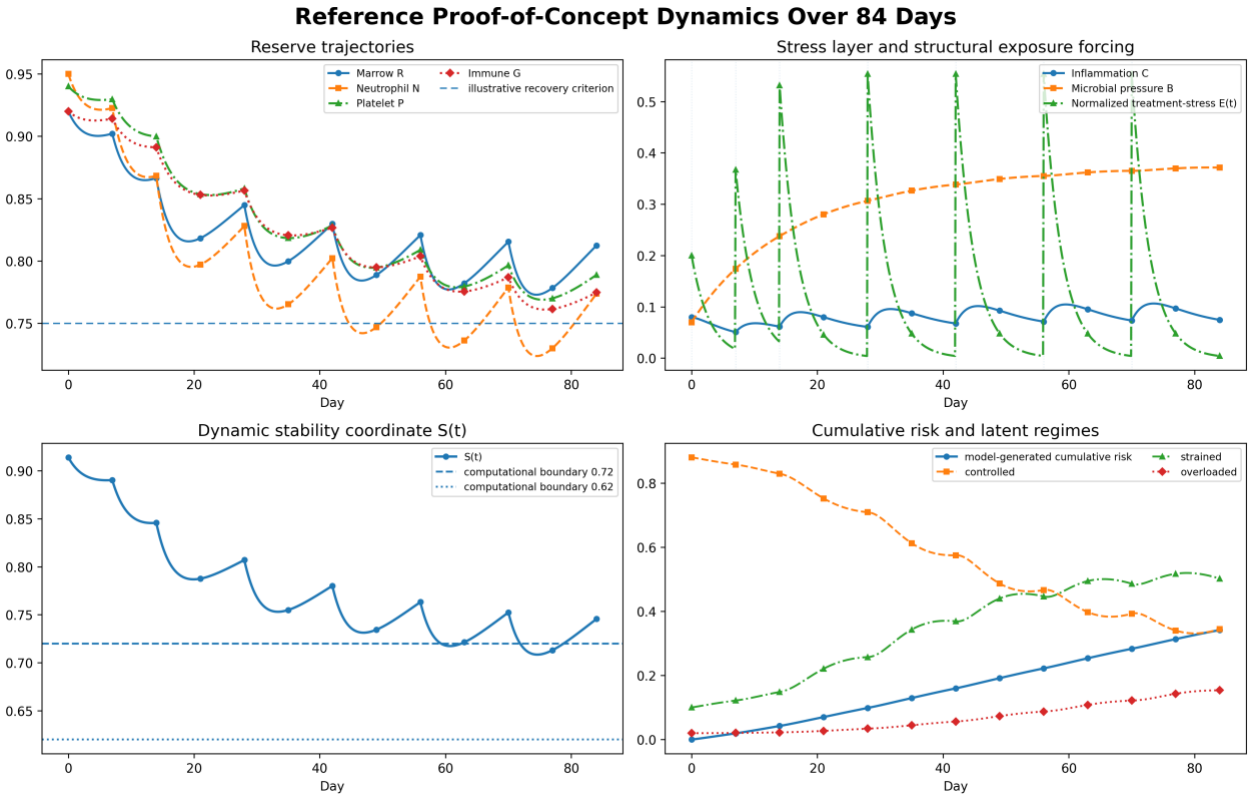


Figure 1. Reference proof-of-concept dynamics over 84 days.

Upper left shows normalized marrow, neutrophil, platelet, and immune-reserve trajectories. Upper right shows inflammatory burden, microbial pressure, and normalized treatment-associated stress forcing. Lower left shows computational stability coordinate with provisional boundaries at $S = 0.72$ and $S = 0.62$. These are model interrogation boundaries, not clinical cutoffs. Lower right shows model-based cumulative infection risk output and latent controlled, strained, and overloaded state probabilities. Repeated trajectory perturbations arise from the declared structural treatment-stress forcing and not be interpreted as observed patient dynamics or a validated pharmacokinetic profile.

3.2. Frailty Dose Intensity Vulnerability Surface

The model-based vulnerability surface showed increasing 84-day infection risk output with increasing frailty and relative dose intensity over the prespecified computational grid (Figure 2). Frailty was varied from 0.10 to 0.90 and relative dose intensity from 0.70 to 1.20 while the remaining model assumptions were held at their reference values. Across this grid, the plotted contours span approximately 0.25-0.55 model-based risk, with the highest values occurring in the region combining greater frailty and greater relative dose intensity.

The surface represents the behaviour of the specified hazard model under parameter perturbation and not be interpreted as an empirically estimated clinical risk surface. In particular, contour values such as 0.25, 0.30, or 0.45 correspond to model-based 84-day risk outputs of 25%, 30%, or 45% under the model, rather than observed infection incidences.

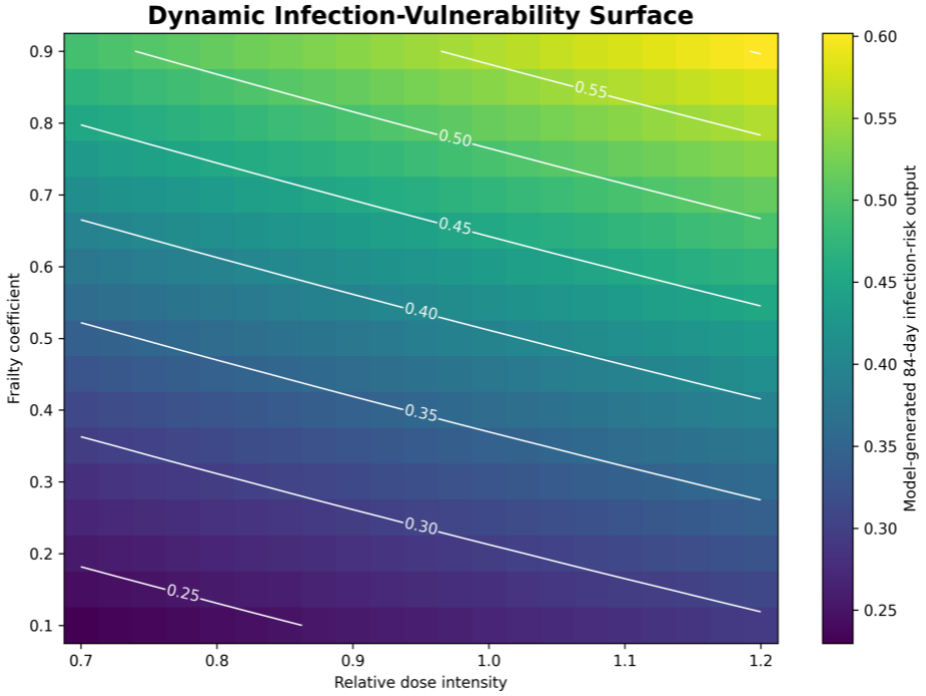


Figure 2. Dynamic infection vulnerability surface.

Model-based 84-day infection risk output across the prespecified frailty (0.10-0.90) and relative dose intensity (0.70-1.20) grid. Contours denote model outputs rather than observed clinical incidence or validated risk thresholds.

3.3. Prespecified Scenario Stress Tests

Prespecified scenario perturbations produced distinct changes in reserve, stability, cumulative risk, and latent-state outputs (Figure 3, Table 1). Relative to the reference configuration, lower dose intensity reduced the model-based risk output from 0.342 to 0.310 and increased neutrophil nadir from 0.724 to 0.774. Growth factor support produced a neutrophil nadir of 0.906 and risk output of 0.268, whereas antimicrobial prophylaxis produced values of 0.731 and 0.295, respectively. Combined adaptation produced the lowest model-based risk among the listed scenarios (0.218) and a neutrophil nadir of 0.950.

Conversely, the high frailty scenario increased model-based risk from 0.342 to 0.492 while changing neutrophil nadir only from 0.724 to 0.719. This separation is a direct consequence of the specified frailty term and hazard equation and is therefore a model implication, not evidence that frailty has this magnitude of clinical effect. Increased marrow injury and increased neutrophil suppression reduced neutrophil nadirs to 0.682 and 0.681 and produced model-based risks of 0.368 and 0.359, respectively.

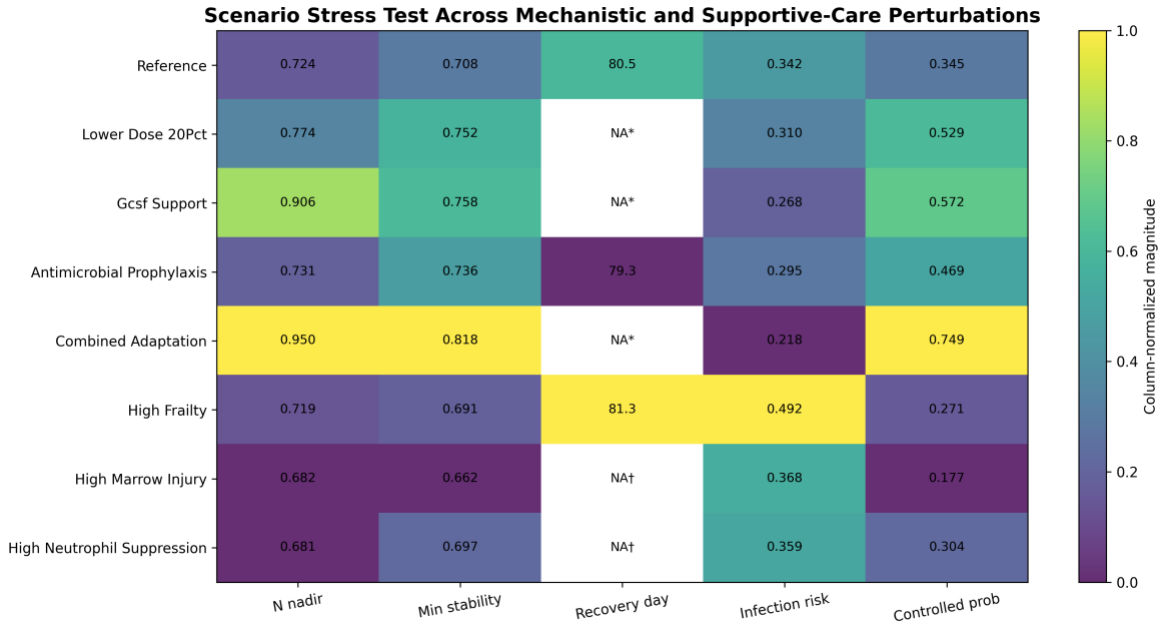


Figure 3. Scenario stress test across mechanistic and supportive care perturbations

Heatmap cells display model-based outputs for neutrophil nadir, minimum stability, recovery day metric, cumulative infection risk output, and final controlled state probability. NA denotes that the predefined recovery day criterion was not applicable or was not reached under the corresponding scenario. It does not denote missing patient data.

Table 1. Model-Based Outputs Under Prespecified Scenario Perturbations

Scenario	N nadir	Min S	Recovery day	Risk output	Controlled probability
Reference	0.724	0.708	80.500	0.342	0.345
Lower dose intensity (20%)	0.774	0.752	NA	0.310	0.529
Growth-factor support	0.906	0.758	NA	0.268	0.572
Antimicrobial prophylaxis	0.731	0.736	79.300	0.295	0.469
Combined adaptation	0.950	0.818	NA	0.218	0.749
High frailty	0.719	0.691	81.300	0.492	0.271
High marrow injury	0.682	0.662	NA	0.368	0.177
High neutrophil suppression	0.681	0.697	NA	0.359	0.304

All entries are model-based outputs under prespecified parameter perturbations and are not observed intervention effects. 'NA' indicates that the predefined recovery day statistic was not applicable or not reached within the corresponding trajectory. It does not represent missing clinical information.

3.4. Global Parameter Sensitivity

Global sensitivity analysis identified strong parameter dependence of the model-based infection risk output (Figure 4). Among the reported PRCCs, the largest positive associations were observed for frailty (0.963), baseline hazard $h_0(0.963)$, microbial input parameter $\lambda_B(0.877)$, neutrophil related hazard coefficient β_N (0.726), and relative dose intensity (0.691). The microbial-clearance parameter clear_B showed a strong negative association (PRCC = -0.763). These coefficients describe monotonic parameter output relationships across the 600 declared Latin hypercube parameter sets. They do not quantify clinical effect sizes, causal associations, or parameter identifiability.

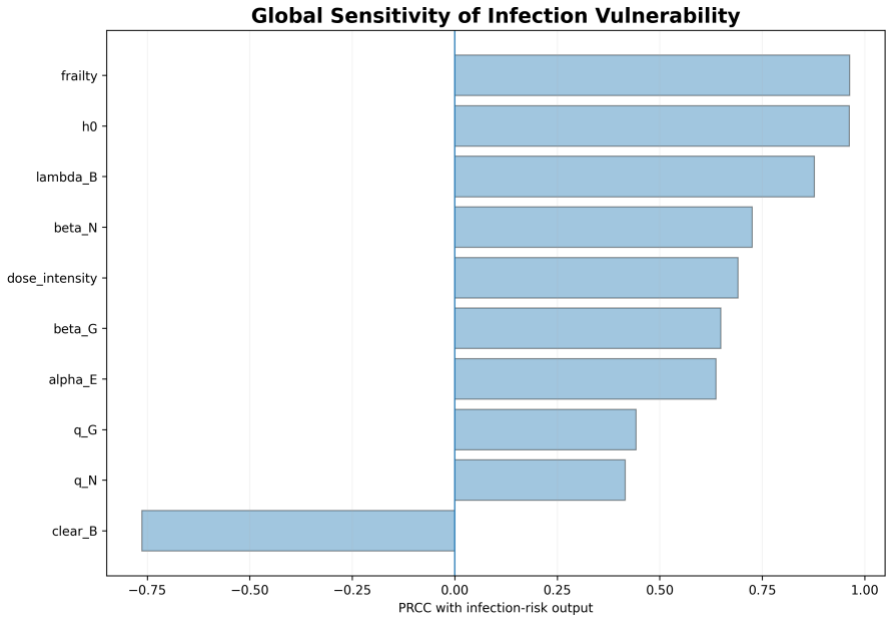


Figure 4. Global sensitivity of model-based infection vulnerability

PRCCs between sampled model parameters and the 84-day model-based infection risk output across 600 Latin hypercube parameter sets. Positive and negative values indicate the direction of the monotonic model association; they are not clinical regression coefficients.

3.5. Stochastic Perturbation Analysis

Across 500 stochastic perturbation runs, the median 84-day model-based infection risk output was 0.349 (34.9%), with a 2.5th-97.5th percentile range of 0.273-0.422 (27.3%-42.2%) (Figure 5). The histogram therefore describes the distribution made by the declared parameter perturbation scheme rather than sampling uncertainty from a patient population. The accompanying neutrophil nadir versus risk scatter plot showed that model-based risk was not determined uniquely by neutrophil nadir within the perturbation ensemble. Variation remained at similar neutrophil reserve values because several additional parameters contribute to the hazard calculation. This pattern is a property of the multivariable model structure and not be interpreted as an observed clinical association.

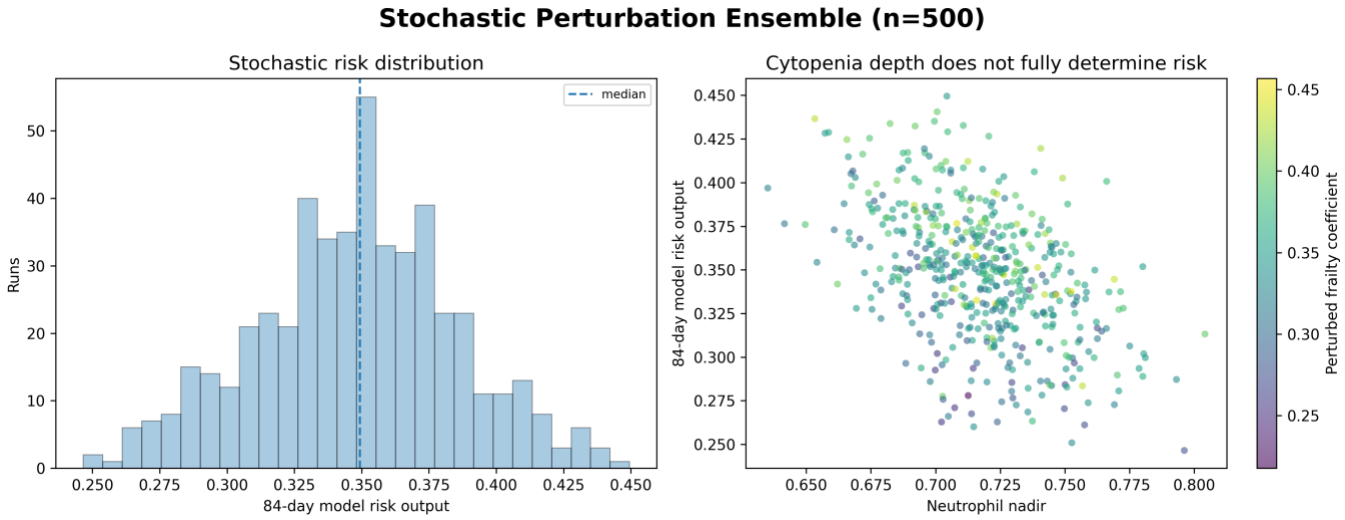


Figure 5. Stochastic uncertainty and separation of neutrophil nadir from cumulative vulnerability.

Left shows distribution of 84-day model-based infection risk outputs, with the median indicated. Right shows relationship between neutrophil nadir and model-based risk across the same perturbation ensemble. The displayed distribution reflects uncertainty under the declared computational perturbations and is not a clinical confidence or prediction interval.

3.6. Latent Regime Dynamics

The three state latent regime representation showed declining controlled state probability and increasing strained and overloaded probabilities over the 84-day reference trajectory (Figure 6). At the end of the simulation, the probabilities were 0.345 controlled, 0.502 strained, and 0.154 overloaded. These quantities are numerical summaries based from the provisional stability dependent transition structure. They are not observed proportions of patients experiencing clinical toxicity states.

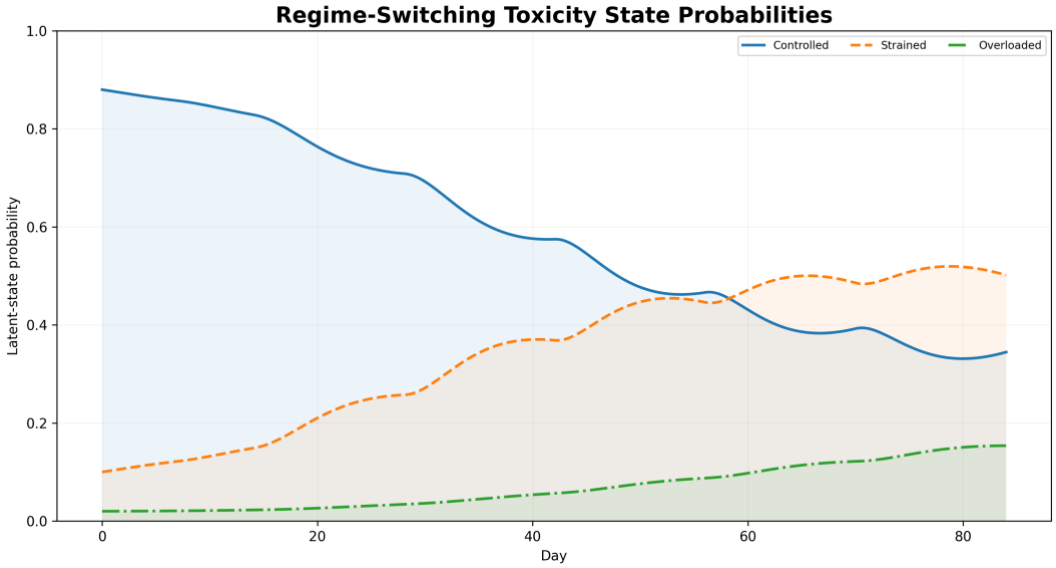


Figure 6. Model-based latent regime probabilities over 84 days

Probabilities of controlled, strained, and overloaded states obtained from the provisional stability dependent regime model. The probabilities summarize the reference model trajectory and do not represent empirically estimated clinical state frequencies.

3.7. Synthetic Structural Model Recovery

In the synthetic mechanism-recovery experiment, the generating dynamic-reserve ODE achieved an AIC of -111.48, compared with -85.95 for the bi-exponential phenomenological model and -67.98 for the 28-day harmonic model. Corresponding Δ AIC values relative to the lowest-AIC model were 0.00, 25.53, and 43.50, respectively (Figure 7). This comparison was intentionally internal. The pseudo-observations were conducted from the mechanistic reference trajectory itself. Consequently, recovery of the model is an internal structural consistency check and does not show superior prediction of clinical data or superiority over alternative clinical models**.**

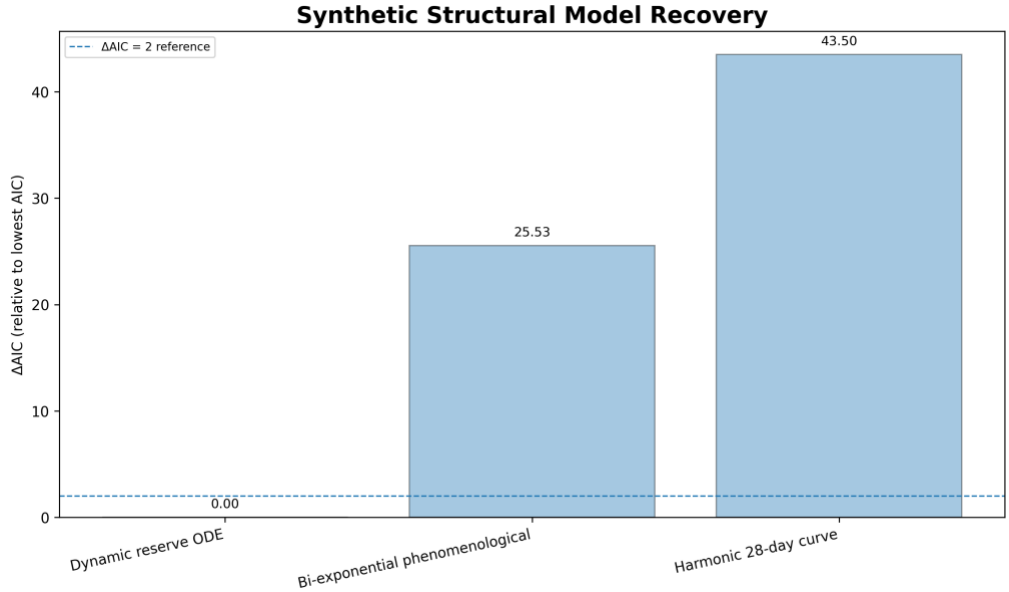


Figure 7. Internal synthetic structural model recovery, pseudo-observations only.

Difference in AIC relative to the lowest-AIC candidate for the dynamic-reserve ODE, bi-exponential phenomenological model, and 28-day harmonic curve. Pseudo-observations were based from the reference mechanistic trajectory. Therefore, the comparison evaluates internal recovery of the generating structure rather than external predictive validity.

3.8. Numerical Robustness

The reference outputs were numerically stable across the tested solver and output grid settings (Figure 8). Relative deviations from the reference configuration were approximately within $\pm 0.002\%$ for the displayed 84-day risk output, neutrophil nadir, and minimum stability. The Table 2 shows that the coarse, reference, and tight settings produced closely similar numerical values. This analysis supports numerical stability of the reported computation under the tested tolerances and grid spacings. It does not establish biological robustness, parameter validity, or clinical reproducibility.

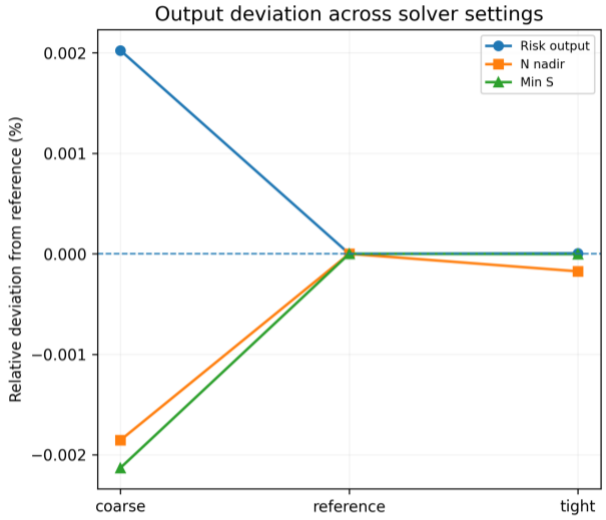


Figure 8. Numerical robustness across solver tolerances.

Relative deviations of selected model outputs from the reference solver configuration and corresponding absolute values. Near zero deviations indicate numerical consistency under the tested computational settings, not clinical validation.

Table 2. Numerical Robustness by Solver Settings

Setting	rtol	atol	dt	Risk	N nadir	Min S
Coarse	1e-06	1e-08	0.20	0.341591	0.723824	0.708468
Reference	1e-08	1e-10	0.10	0.341584	0.723838	0.708483
Tight	1e-10	1e-12	0.05	0.341584	0.723837	0.708483

Risk denotes the 84-day model-based infection risk output. N nadir denotes the minimum normalized neutrophil reserve, and Min S denotes minimum computational stability. The close agreement across coarse, reference, and tight numerical settings that the reported outputs are numerically stable under the tested solver tolerances and output grid resolutions. This analysis assesses numerical robustness only and does not establish biological or clinical validity.

3.9. Structural Exposure Schedule Sensitivity

The model showed substantial dependence on the assumed structural treatment-stress schedule (Figure 9). Under the reference q14 configuration, neutrophil nadir, minimum stability, and 84-day model-based infection risk output were 0.724, 0.708, and 0.342, respectively. Weekly-after-step-up forcing changed these outputs to 0.555, 0.552, and 0.461. Under this schedule, the trajectory spent 34.35 model days below the $S = 0.62$ computational boundary. In contrast, q21-after-step-up forcing produced a neutrophil nadir of 0.790, minimum stability of 0.765, and risk output of 0.306. These differences supports that exposure timing is a consequential structural assumption of the model. They do not establish that one clinical dosing regimen is safer than another, because the forcing schedules are normalized computational inputs rather than validated drug-specific dosing or pharmacokinetic profiles.

Table 3. Structural Exposure Schedule Sensitivity

Schedule	N nadir	Min S	Risk Output
reference_q14	0.724	0.708	0.342
weekly_after_stepup	0.555	0.552	0.461
q21_after_stepup	0.790	0.765	0.306

Values are model-based outputs under alternative normalized treatment stress forcing schedules. The schedules are structural sensitivity experiments and do not represent comparative clinical regimen outcomes.

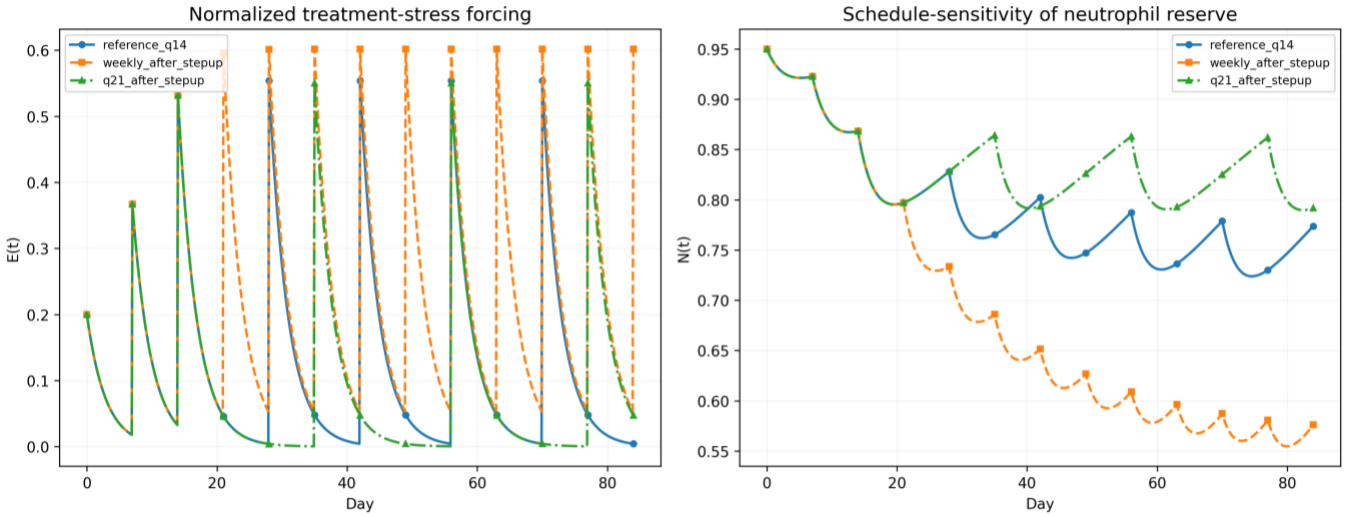


Figure 9. Structural exposure schedule sensitivity.

Left shows normalized treatment associated stress forcing under the reference q14, weekly-after-step-up, and q21-after-step-up structural schedules. Right shows corresponding model-based neutrophil reserve trajectories. These schedules are computational forcing assumptions and not be interpreted as validated dosing or pharmacokinetic profiles.

4. DISCUSSION

This study developed a literature-parameterized dynamic reserve stress hazard model to examine whether cytopenia depth and broader infection vulnerability can diverge over time during bispecific antibody therapy. Within the declared reference configuration, the model conducted an 84-day neutrophil nadir of 0.724, minimum computational stability of 0.708, and cumulative model-based infection risk output of 34.2%. These values are interpreted as outputs of the specified equations and provisional parameterization rather than estimates of clinical incidence, treatment effect, or validated risk thresholds. The principal contribution of the model is not prediction of patient outcomes, but an explicit mathematical representation in which marrow reserve, circulating cell reserves, immune competence, inflammatory burden, microbial pressure, treatment associated stress, and cumulative hazard evolve jointly over time.

A central model implication was that neutrophil depth alone did not uniquely determine cumulative vulnerability. Under the high frailty scenario, the neutrophil nadir changed only from 0.724 in the reference configuration to 0.719, whereas the 84-day model-based risk output increased from 34.2% to 49.2%. Conversely, increased marrow injury and increased neutrophil suppression produced deeper neutrophil nadirs of 0.682 and 0.681 but lower risk outputs of 36.8% and 35.9%, respectively, than the high frailty scenario. These contrasts arise from the specified multicomponent hazard structure, in which neutrophil reserve is only one contributor alongside immune competence, inflammatory burden, microbial pressure, and frailty. They therefore support the mathematical objective of demonstrating that similar or greater cytopenia depth can coexist with different model-based vulnerability histories. They do not establish that frailty or any other component has the same quantitative effect in patients.

The sensitivity and uncertainty analyses further showed that this behaviour depends materially on the assumed model structure and parameterization. Across 600 Latin hypercube samples, the largest reported positive PRCCs for the 84-day risk output were associated with frailty (0.963), baseline hazard $h_0(0.963)$, microbial input $\lambda_B(0.877)$, the neutrophil-related hazard coefficient $\beta_N(0.726)$, and relative dose intensity (0.691), whereas microbial clearance clear_B showed a strong negative association (-0.763). Across 500 stochastic perturbations, median model-based risk was 34.9%, with a 2.5^{th}-97.5^{th} percentile range of 27.3%-42.2%. These analyses supports parameter dependence and propagation of the declared computational uncertainty. They are not interpreted as clinical effect estimates, confidence intervals, or evidence that the corresponding parameters are identifiable from patient data.

The treatment stress schedule audit identified an additional source of structural uncertainty. The reference q14 forcing produced a neutrophil nadir of 0.724, minimum stability of 0.708, and risk output of 0.342, whereas weekly-after-step-up forcing produced corresponding values of 0.555, 0.552, and 0.461. The q21-after-step-up configuration produced values of 0.790, 0.765, and 0.306. The weekly structural schedule also shown 34.35 model days below the $S = 0.62$ computational boundary. These differences show that the simulated trajectories are sensitive to the timing of the assumed forcing. This finding is important because $E(t)$ is intentionally a normalized treatment associated stress input rather than serum drug concentration, receptor occupancy, or a validated pharmacokinetic profile. Accordingly, the schedule comparison is not interpreted as evidence that a weekly clinical regimen is more hazardous

or that a 21-day regimen is safer. Instead, it identifies treatment stress timing as a structural assumption that requires drug specific justification or calibration before clinical application.

The latent regime representation provided a complementary summary of the reference trajectory. By day 84, the model-based probabilities were 34.5% controlled, 50.2% strained, and 15.4% overloaded. These quantities translate the continuous stability trajectory into a simplified time varying state representation, but the transition intensities were provisional and the executable implementation used numerical updating with clipping and renormalization. The resulting probabilities are therefore best regarded as descriptive latent state summaries rather than empirically estimated clinical state prevalence or exact patient transition probabilities. Similarly, the computational stability boundaries were introduced for mathematical interrogation of the trajectory and not interpreted as clinically validated cut-offs.

Several analyses support the internal computational consistency of the model, but their evidential scope is deliberately limited. The reference outputs changed only minimally across the tested coarse, reference, and tight numerical settings, indicating that the reported values were not materially driven by the selected solver tolerance or output grid resolution. In the synthetic mechanism recovery experiment, the generating dynamic reserve ODE had an AIC of -111.48, compared with -85.95 for the bi-exponential model and -67.98 for the harmonic model. However, because the pseudo-observations were conducted from the mechanistic reference trajectory itself, this comparison is an internal recovery check by design. It does not show superior prediction, model selection, or clinical validity on independent patient data.

The analyses support the model primarily as a hypothesis mathematical proof-of-concept**.** The reference trajectory provides the behaviour of one declared configuration. The scenario analysis illustrates how mechanistic assumptions can separate cytopenia depth from cumulative vulnerability. The risk surface examines joint frailty dose intensity dependence. Sensitivity and stochastic analyses expose parameter dependence, and numerical robustness tests computational stability. None of these analyses converts the model into a validated clinical risk tool. Patient level longitudinal data would be required to estimate kinetic parameters, assess structural and practical identifiability, replace or calibrate the treatment stress forcing where appropriate, evaluate competing clinical outcomes, and test predictive performance in independent cohorts. Until those steps are completed, the numerical values reported here remain interpreted as reproducible model outputs rather than patient-specific predictions or treatment recommendations.

5. LIMITATIONS

This study is a literature-parameterized computational proof-of-concept and was not calibrated or validated using patient level longitudinal data. Accordingly, the reported risk outputs, latent regime probabilities, stability boundaries, scenario contrasts, and uncertainty distributions are model-based quantities rather than clinical incidences, validated thresholds, or treatment effect estimates, and parameter identifiability has not been established. The biological representation is intentionally simplified, with immune competence represented by a normalized state, microbial pressure treated phenomenologically, and infection vulnerability represented through cumulative hazard. Treatment exposure is represented by a normalized structural stress forcing rather than a validated drug specific pharmacodynamic profile. The observed schedule sensitivity therefore reflects dependence on this

modelling assumption and cannot be interpreted as a comparison of clinical dosing regimens. In addition, the latent regime probabilities rely on provisional transition intensities and numerical Euler updating with clipping and renormalization, while the stochastic, sensitivity, numerical robustness, and synthetic mechanism recovery analyses assess internal model behavior rather than external predictive validity. Patient level calibration, identifiability assessment, drug specific exposure modelling, and independent external validation are required before clinical interpretation or decision use.

6. CONCLUSION

This literature-parameterized computational proof-of-concept provides a reproducible dynamic reserve stress hazard model for examining the relationship between cytopenia recovery and broader infection vulnerability during bispecific antibody therapy. Under the declared model assumptions, the analyses showed that similar neutrophil nadirs can coexist with substantially different model-based vulnerability, while sensitivity, stochastic perturbation, scenario, numerical robustness, and exposure schedule analyses showed that these outputs depend materially on the specified parameters and structural forcing assumptions. The model is interpreted as a hypothesis mathematical model for interrogating dynamic mechanisms and informing future longitudinal data collection and calibration, rather than as a source of clinical incidence estimates, treatment effects, validated thresholds, or treatment recommendations. Patient level calibration, parameter identifiability assessment, drug specific exposure modelling where appropriate, and independent external validation are required before clinical prediction or decision use.

7. DATA AND CODE AVAILABILITY

The computational code, model inputs, and model-based outputs supporting this study are provided in the accompanying public repository. The executable workflow reproduces the reference trajectory, prespecified scenario analyses, dynamic risk surface, 600-set Latin-hypercube (PRCC) sensitivity analysis, local elasticity analysis, 500-run stochastic perturbation analysis, numerical robustness assessment, synthetic AIC/BIC mechanism recovery experiment, and structural exposure schedule sensitivity analysis showed in the manuscript. Corresponding model-based results are provided as structured CSV/JSON outputs together with the manuscript figures. No patient level clinical data were used, analyzed, or included in the repository. All numerical results reported here arise from the declared literature-parameterized computational model and its specified assumptions.

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